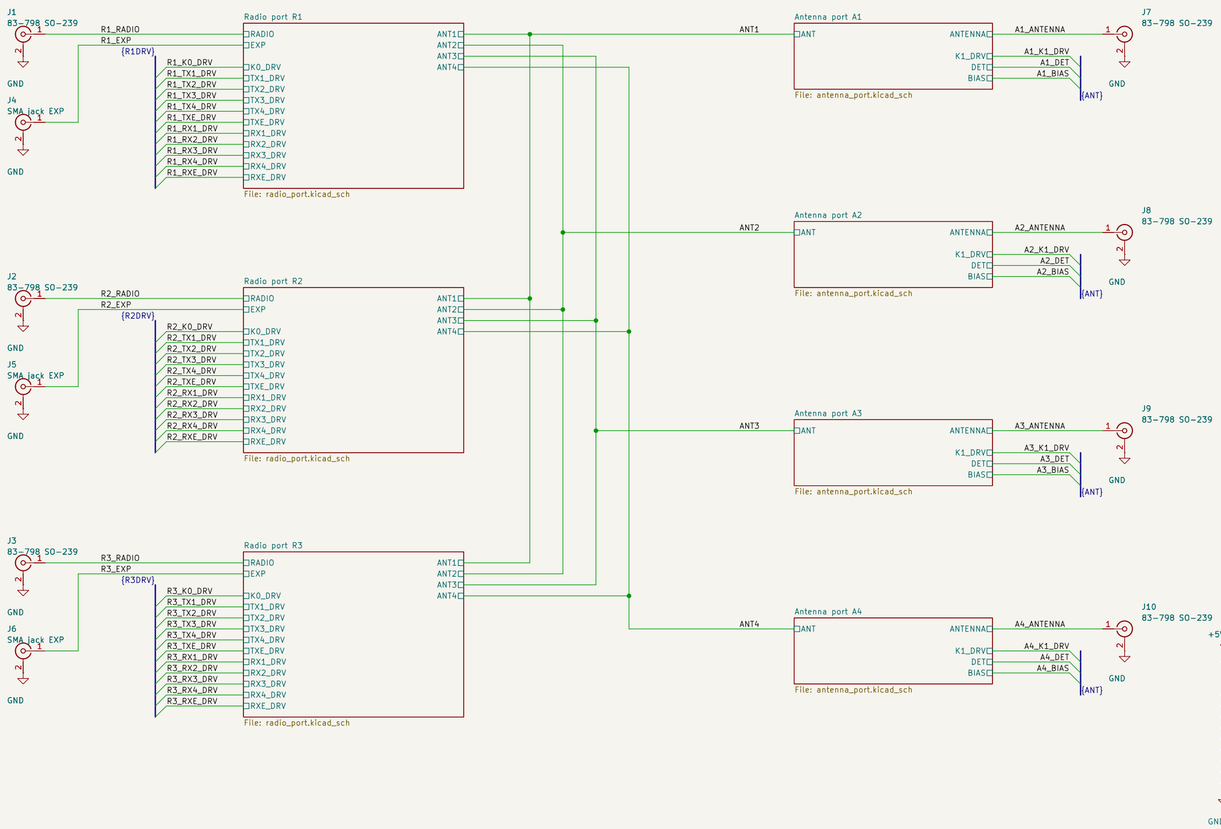


Shack Relay Matrix, unit RF board, architecture rev C. One flat board in a milled 6082-T6 case.
All connectors are on this sheet: radio SO-239 J1 to J3 and EXP SMA J4 to J6 (left), antenna SO-239 J7 to J10
(right), power J11 to J13. Unit 1: 3 radios x 4 antennas: A1 G5RV, A2 RX loop (RX only), A3 and A4 spare.
Crossbar: the four vertical wires are the antenna columns ANT1 to ANT4 (L2). Each radio sheet holds
a TX row and an RX row (L1) with one relay per column, 37 x RT1: 3 K0, 15 TX, 15 RX, 4 K1; unit 1 fits 34 (no TX relay at A2, D-103).
Coil drivers silt next to each relay; Rn_*_DRV and An_K1_DRV are 3.3 V gate inputs from the IO expanders sheet.
J14 goes to the controller board (unit 1) or takes the EXP ribbon from the unit 1 controller board (unit 2), D-51, D-53;
each J14 signal passes the J14 filter sheet: series link, C footprint and test point (D-122, D-120).
The RF board is the same in both units. R1_K0 follows PTT in hardware (TX_ANY, R51, D-58); R2, R3 K0 from U903 (D-63).
Control lines as buses (D-60): R1DRV to R3DRV (coil drives per radio), ANT (K1 drive, detector, bias feed per antenna port),
CTRL (I2C, EXP_INT, RLY_EN). One bus segment at each block; the bus alias joins them, each entry wire carries its net name.
Test: ERC; netlist check that each ANTx joins 6 row relays and one antenna port; logic verification (verify_logic.py, M-08).
Target: 50 ohm RF path, 100 W HF at SWR 3 (173 V peak, 2.45 A rms), PD3 outdoor box.

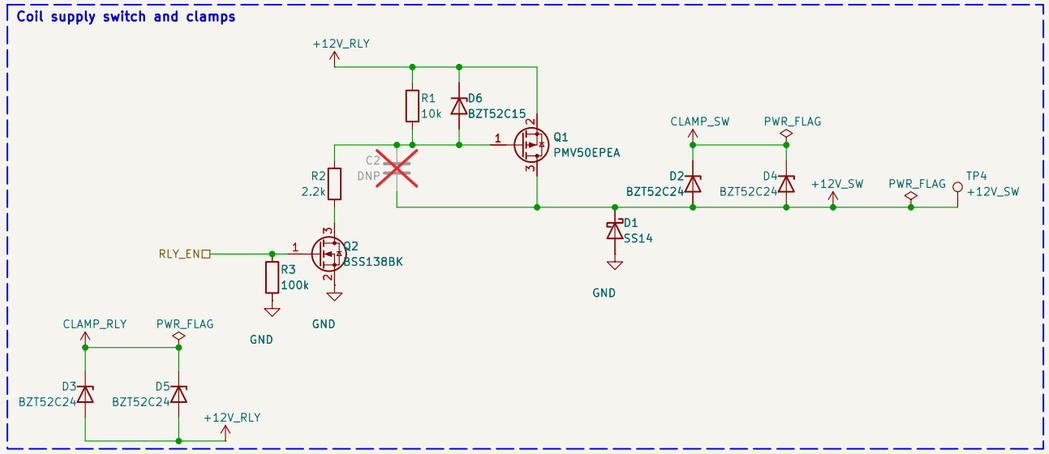
JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

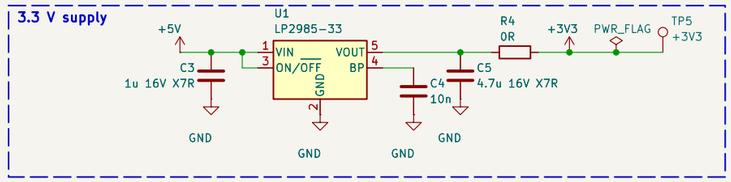
PE1RGU - Shack Relay Matrix

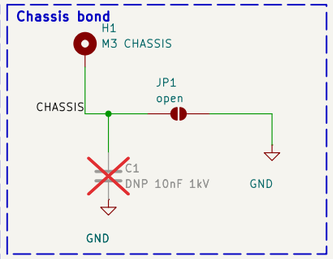
Sheet: /
File: rf-board.kicad_sch

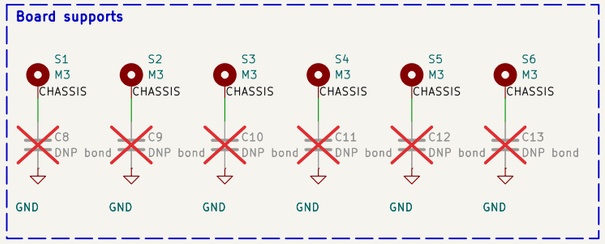
Title: Unit RF board: overview, connectors and crossbar

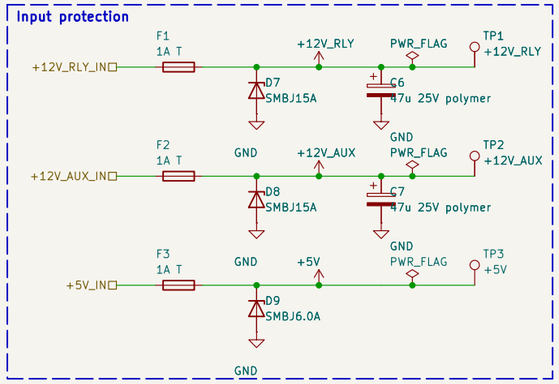
Size: A2	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 1/13











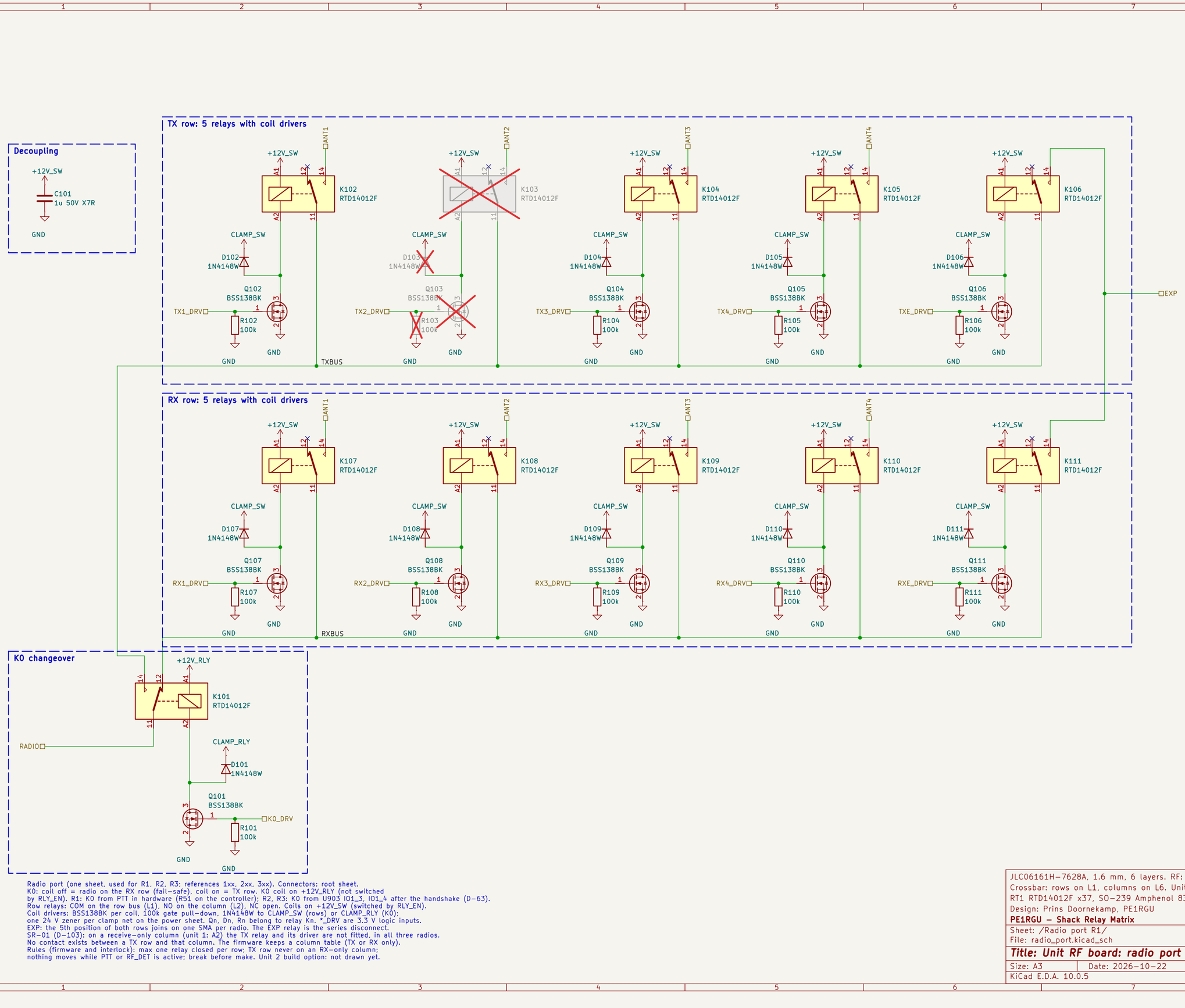
Power: the XT connectors sit on the root sheet. Each input: 1 A time-lag fuse, then a TVS to GND (D-45).
The TVS clamps transients and conducts on reverse polarity, so the fuse opens. D6 limits Q1 VGS to 15 V.
+12V_RLY: 12 V +-5 % (RT1 coil range 10.2 to 13.2 V). +12V_SW: +12V_RLY switched by Q1 when RLY_EN is high.
RLY_EN is active-high with R3 pull-down: no signal = relays off (D-39). C2 not fitted (soft start option).
Clamp nets: each coil diode goes to CLAMP_SW (rows, K1) or CLAMP_RLY (K0). Zeners D2 // D4 and D3 // D5 (24 V) return
the coil energy to the rail: drain peak about 37.5 V. Two in parallel: one open zener does not remove the clamp.
+3V3: U1 LP2985-33 from +5V, for the expanders and the I2C bus. R4 0R: lift to measure the 3.3 V load (D-120).
Bulk C6, C7 47 uF polymer at the 12 V entries (D-129). Test points TP1 to TP5 on every rail; the fuses F1 to F3
serve as the current links of the input rails (D-120).
Mounting holes not grounded; H1 is the only bond (JP1 open, C1 not fitted). The case bonds to the RVS earth bar at one point only.
Supports S1 to S6: plated, on CHASSIS through the standoffs; C8 to C13 are bond options to GND, not fitted (D-132).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Power, coil supply and 3V3/
File: power.kicad_sch

Title: Unit RF board: power, coil supply, 3V3 and chassis bond

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 2/13



Radio port (one sheet, used for R1, R2, R3; references 1xx, 2xx, 3xx). Connectors: root sheet.
K0: coil off = radio on the RX row (fail-safe), coil on = TX row. K0 coil on +12V_RLY (not switched by RLY_EN). R1: K0 from PTT in hardware (R51 on the controller); R2, R3: K0 from U903 IO1_3, IO1_4 after the handshake (D-63).
Row relays: COM on the row bus (L1), NO on the column (L2), NC open. Coils on +12V_SW (switched by RLY_EN).
Coil drivers: BSS138BK per coil, 100k gate pull-down, 1N4148W to CLAMP_SW (rows) or CLAMP_RLY (K0); one 24 V zener per clamp net on the power sheet. Qn, Dn, Rn belong to relay Kn. *_DRV are 3.3 V logic inputs.
EXP: the 5th position of both rows joins on one SMA per radio. The EXP relay is the series disconnect.
SR-01 (D-103): on a receive-only column (unit 1: A2) the TX relay and its driver are not fitted, in all three radios.
No contact exists between a TX row and that column. The firmware keeps a column table (TX or RX only).
Rules (firmware and interlock): max one relay closed per row; TX row never on an RX-only column; nothing moves while PTT or RF_DET is active; break before make. Unit 2 build option: not drawn yet.

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

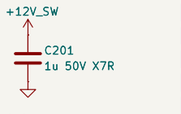
PE1RGU — Shack Relay Matrix

Sheet: /Radio port R1/
File: radio_port.kicad_sch

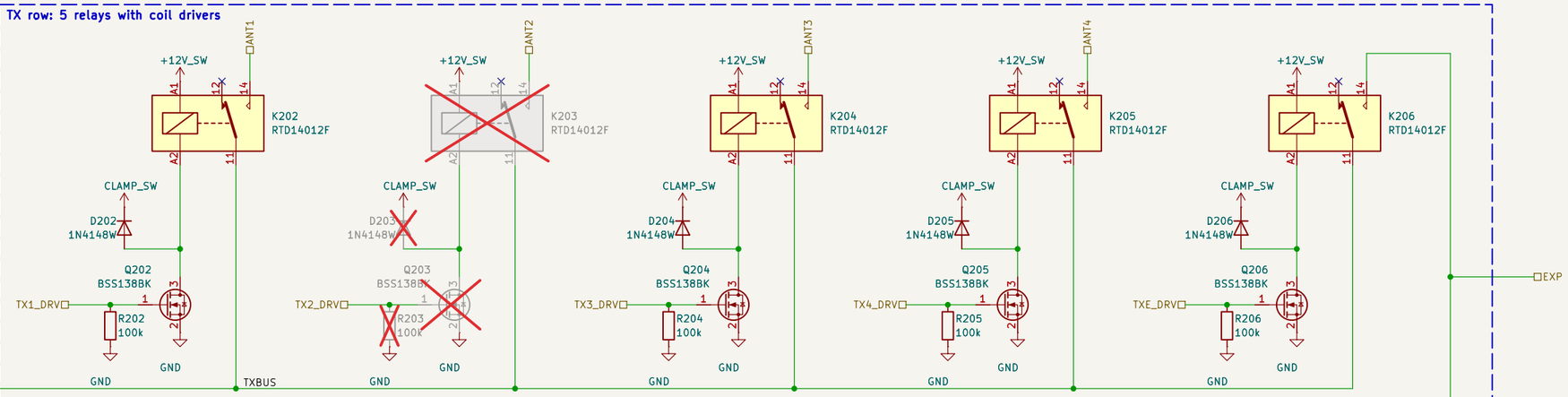
Title: Unit RF board: radio port (K0, TX/RX rows, EXP, drivers)

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 3/13

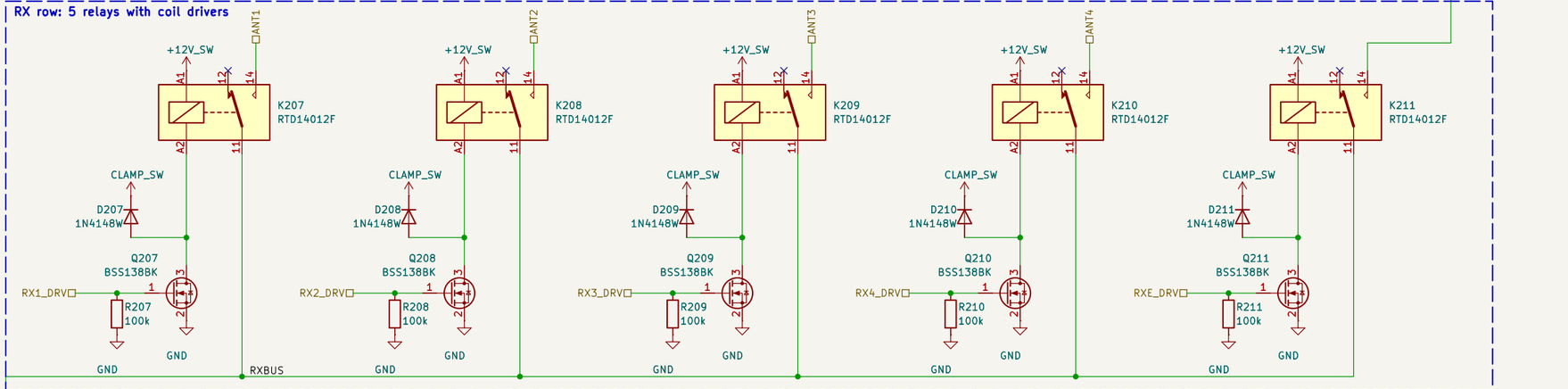
Decoupling



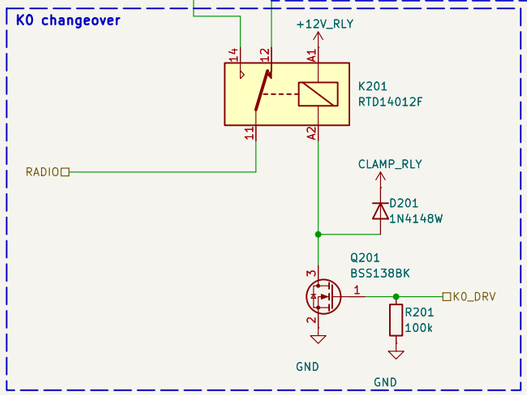
TX row: 5 relays with coil drivers



RX row: 5 relays with coil drivers



K0 changeover



Radio port (one sheet, used for R1, R2, R3; references 1xx, 2xx, 3xx). Connectors: root sheet.
K0: coil off = radio on the RX row (fail-safe), coil on = TX row. K0 coil on +12V_RLY (not switched
by RLY_EN). R1: K0 from PTT in hardware (R51 on the controller); R2, R3: K0 from U903 IO1_3, IO1_4 after the handshake (D-63).
Row relays: COM on the row bus (L1), NO on the column (L2), NC open. Coils on +12V_SW (switched by RLY_EN).
Coil drivers: BSS138BK per coil, 100k gate pull-down, 1N4148W to CLAMP_SW (rows) or CLAMP_RLY (K0);
one 24 V zener per clamp net on the power sheet. Qn, Dn, Rn belong to relay Kn. *_DRV are 3.3 V logic inputs.
EXP: the 5th position of both rows joins on one SMA per radio. The EXP relay is the series disconnect.
SR-01 (D-103): on a receive-only column (unit 1: A2) the TX relay and its driver are not fitted, in all three radios.
No contact exists between a TX row and that column. The firmware keeps a column table (TX or RX only).
Rules (firmware and interlock): max one relay closed per row; TX row never on an RX-only column;
nothing moves while PTT or RF_DET is active; break before make. Unit 2 build option: not drawn yet.

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

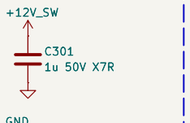
PE1RGU — Shack Relay Matrix

Sheet: /Radio port R2/
File: radio_port.kicad_sch

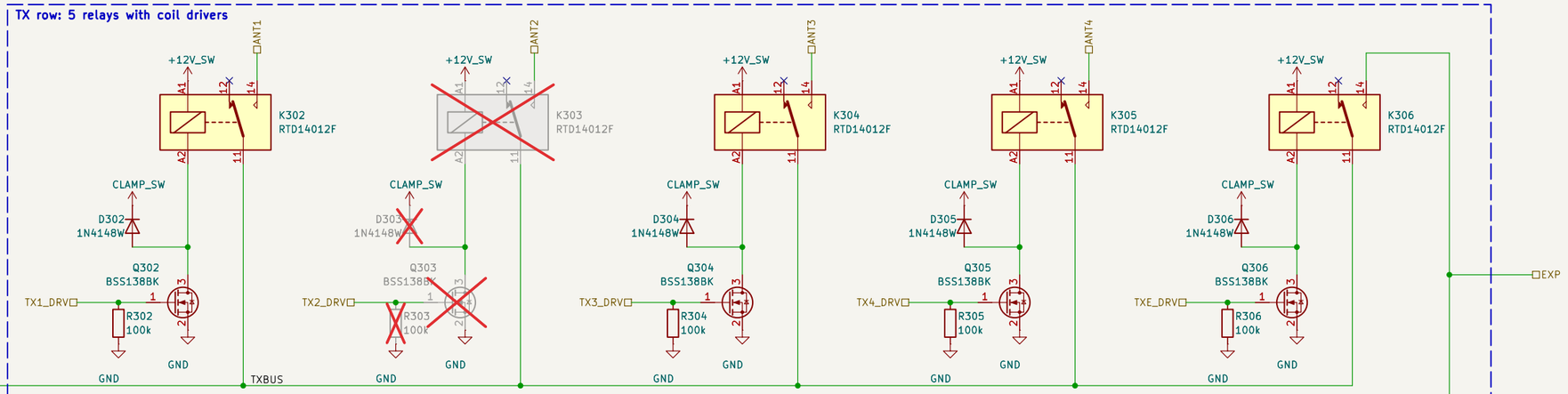
Title: Unit RF board: radio port (K0, TX/RX rows, EXP, drivers)

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 4/13

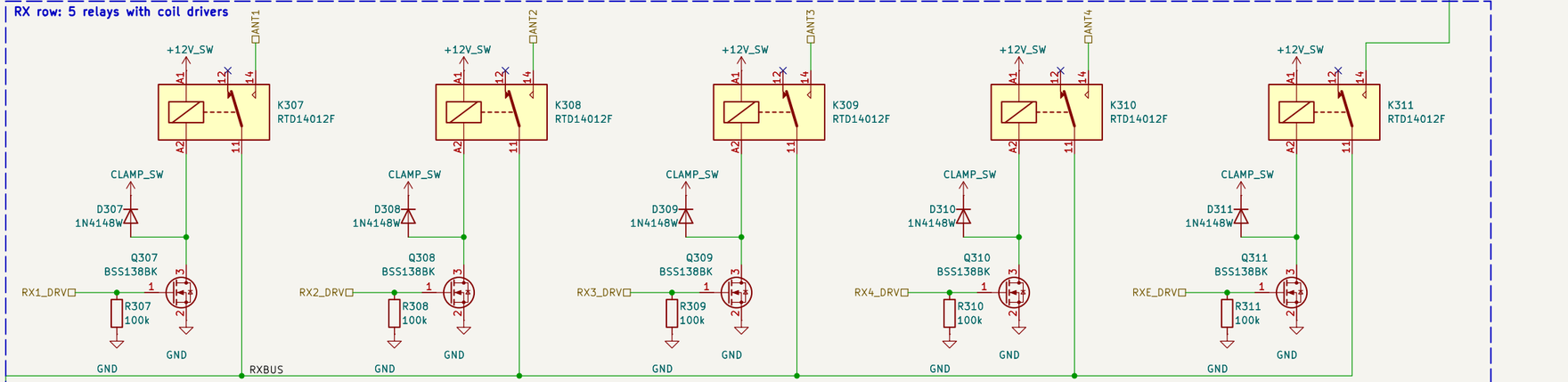
Decoupling



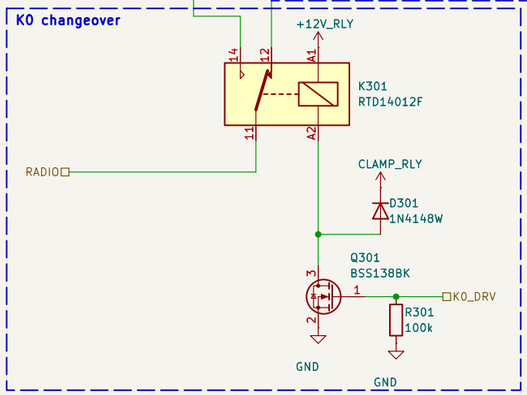
TX row: 5 relays with coil drivers



RX row: 5 relays with coil drivers



K0 changeover



Radio port (one sheet, used for R1, R2, R3; references 1xx, 2xx, 3xx). Connectors: root sheet.
K0: coil off = radio on the RX row (fail-safe), coil on = TX row. K0 coil on +12V_RLY (not switched
by RLY_EN). R1: K0 from PTT in hardware (R51 on the controller); R2, R3: K0 from U903 IO1_3, IO1_4 after the handshake (D-63).
Row relays: COM on the row bus (L1), NO on the column (L2), NC open. Coils on +12V_SW (switched by RLY_EN).
Coil drivers: BSS138BK per coil, 100k gate pull-down, 1N4148W to CLAMP_SW (rows) or CLAMP_RLY (K0);
one 24 V zener per clamp net on the power sheet. Qn, Dn, Rn belong to relay Kn. *_DRV are 3.3 V logic inputs.
EXP: the 5th position of both rows joins on one SMA per radio. The EXP relay is the series disconnect.
SR-01 (D-103): on a receive-only column (unit 1: A2) the TX relay and its driver are not fitted, in all three radios.
No contact exists between a TX row and that column. The firmware keeps a column table (TX or RX only).
Rules (firmware and interlock): max one relay closed per row; TX row never on an RX-only column;
nothing moves while PTT or RF_DET is active; break before make. Unit 2 build option: not drawn yet.

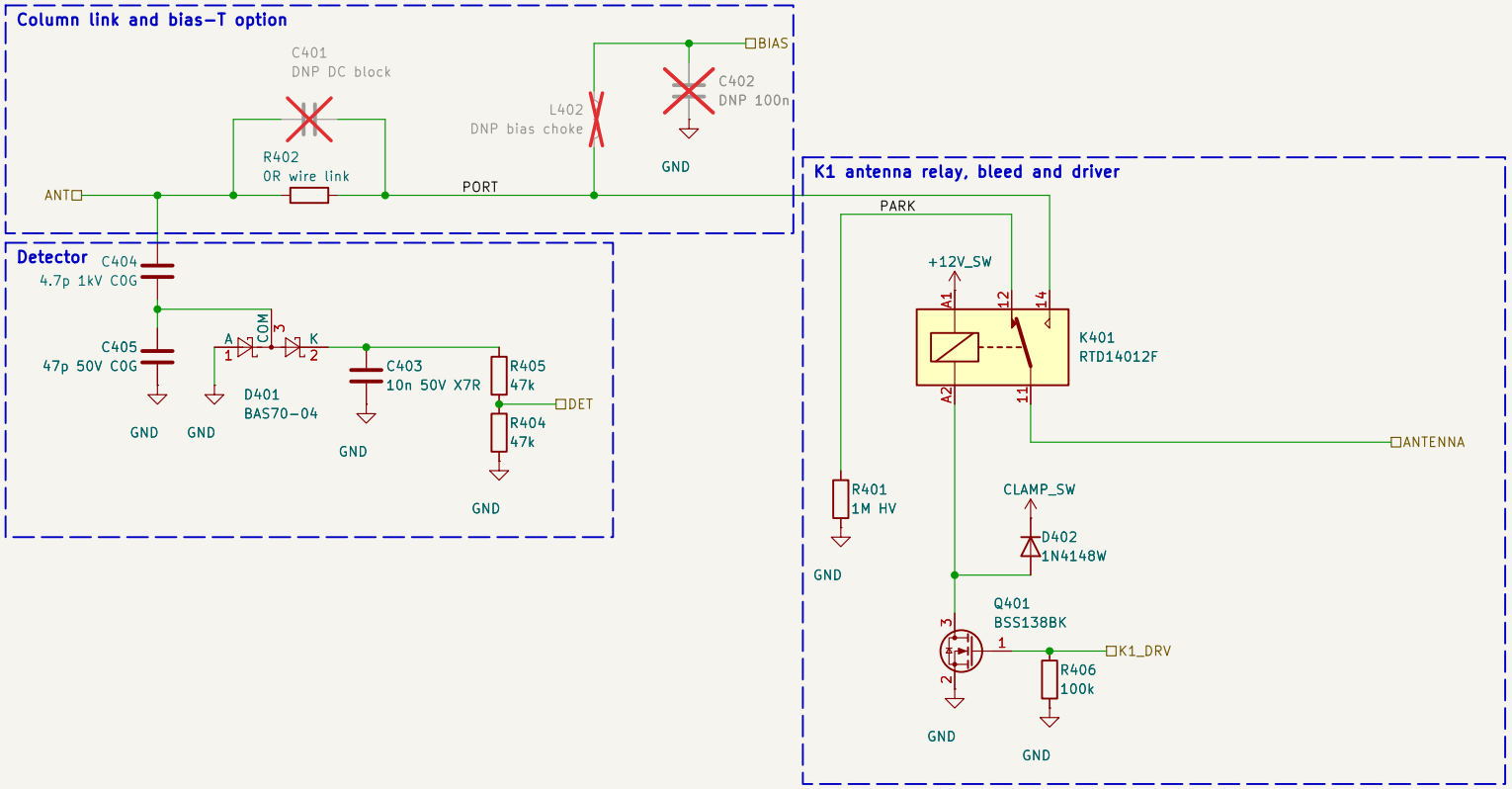
JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

PE1RGU — Shack Relay Matrix

Sheet: /Radio port R3/
File: radio_port.kicad_sch

Title: Unit RF board: radio port (K0, TX/RX rows, EXP, drivers)

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 5/13



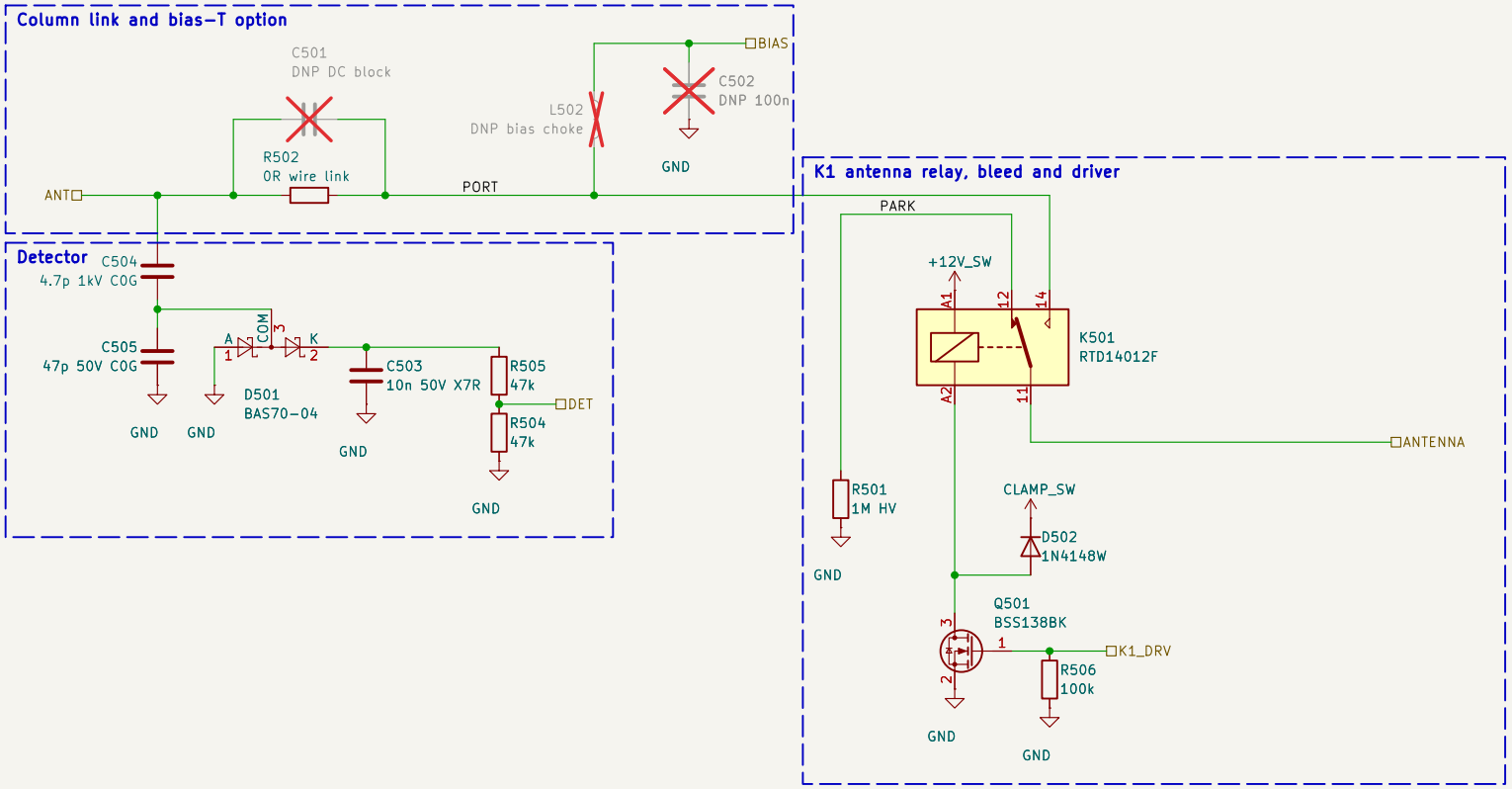
Antenna port (one sheet, used for A1 to A4; references 4xx to 7xx). SO-239: root sheet.
K1: coil off = antenna parked (COM to NC, 1M bleed to GND), coil on = antenna on the column.
No RF choke on the bleed: a low-impedance path would take surge current through K1 and the board.
Surge: Comet CS-400R at the RVS box entry. No GDT on the board (D-114).
Bias-T option (A2 RX loop): fit C1 and L2, C2, remove R2. The feed sits on the matrix side of K1,
so the bias supply is open, not loaded, while K1 is parked.
Detector: capacitive tap C4/C5 (about 1:12), BAS70-04 doubler D1 (clamps negative pulses), hold C3, R5/R4 divide by 2.
K1 driver: Q1 BSS138BK, R6 100k gate pull-down, D2 1N4148W to CLAMP_SW. Values marked TBD are set with the part choices (design-log.md).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Antenna port A1/
File: antenna_port.kicad_sch

Title: Unit RF board: antenna port (K1, driver, detector, bias-T)

Size: A4	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 6/13



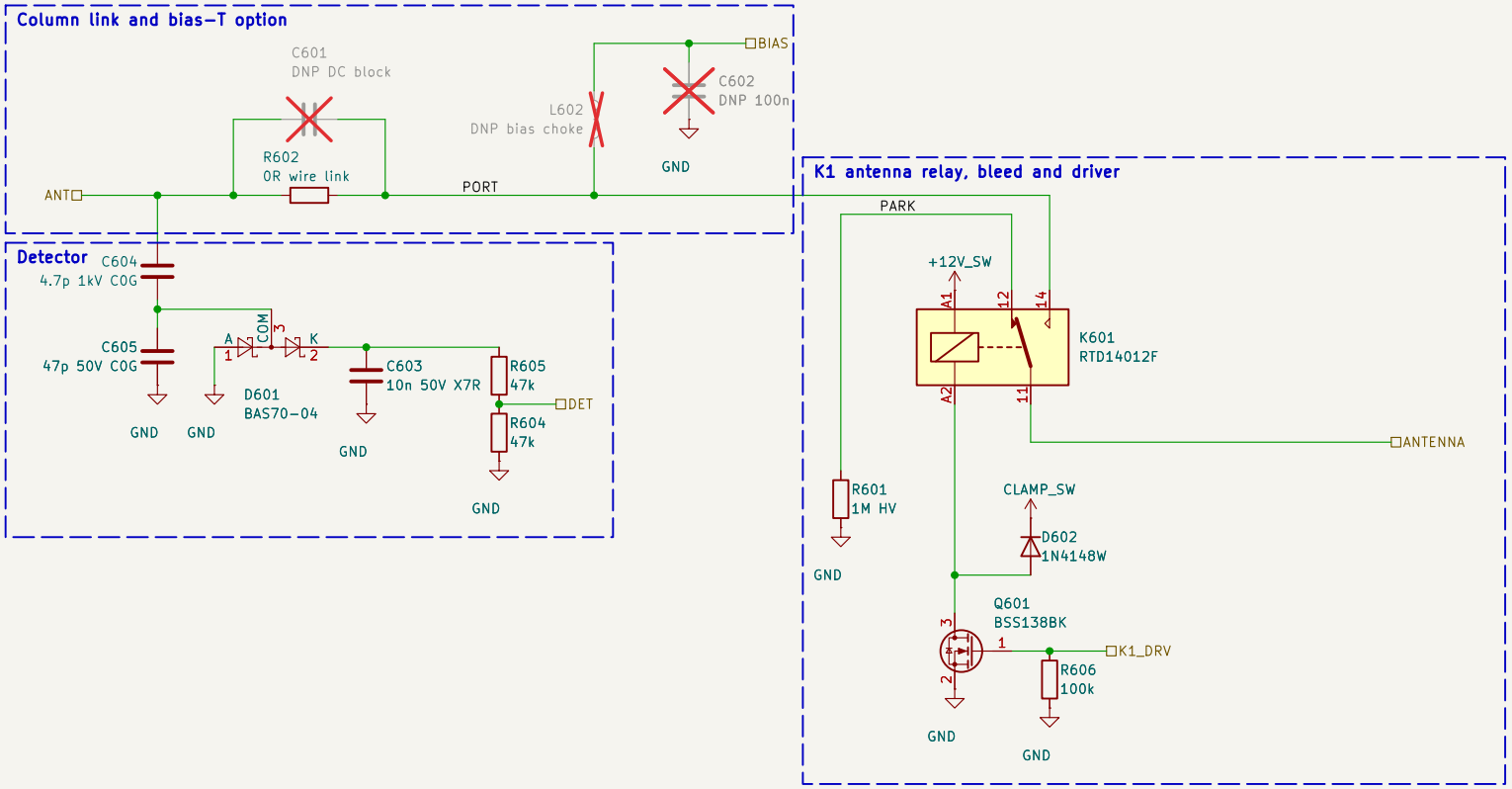
Antenna port (one sheet, used for A1 to A4; references 4xx to 7xx). SO-239: root sheet.
K1: coil off = antenna parked (COM to NC, 1M bleed to GND), coil on = antenna on the column.
No RF choke on the bleed: a low-impedance path would take surge current through K1 and the board.
Surge: Comet CS-400R at the RVS box entry. No GDT on the board (D-114).
Bias-T option (A2 RX loop): fit C1 and L2, C2, remove R2. The feed sits on the matrix side of K1,
so the bias supply is open, not loaded, while K1 is parked.
Detector: capacitive tap C4/C5 (about 1:12), BAS70-04 doubler D1 (clamps negative pulses), hold C3, R5/R4 divide by 2.
K1 driver: Q1 BSS138BK, R6 100k gate pull-down, D2 1N4148W to CLAMP_SW. Values marked TBD are set with the part choices (design-log.md).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Antenna port A2/
File: antenna_port.kicad_sch

Title: Unit RF board: antenna port (K1, driver, detector, bias-T)

Size: A4	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 7/13



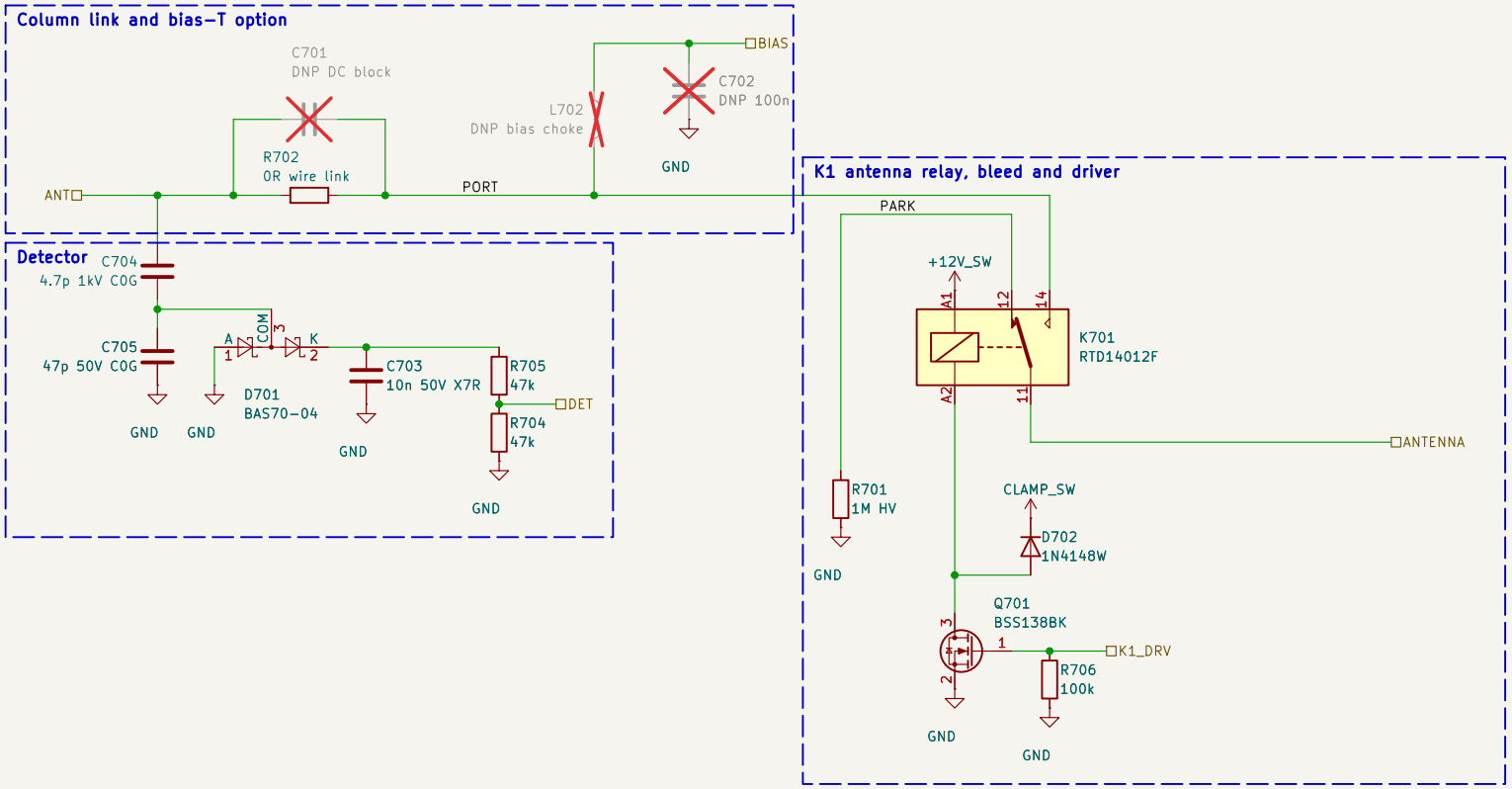
Antenna port (one sheet, used for A1 to A4; references 4xx to 7xx). SO-239: root sheet.
K1: coil off = antenna parked (COM to NC, 1M bleed to GND), coil on = antenna on the column.
No RF choke on the bleed: a low-impedance path would take surge current through K1 and the board.
Surge: Comet CS-400R at the RVS box entry. No GDT on the board (D-114).
Bias-T option (A2 RX loop): fit C1 and L2, C2, remove R2. The feed sits on the matrix side of K1,
so the bias supply is open, not loaded, while K1 is parked.
Detector: capacitive tap C4/C5 (about 1:12), BAS70-04 doubler D1 (clamps negative pulses), hold C3, R5/R4 divide by 2.
K1 driver: Q1 BSS138BK, R6 100k gate pull-down, D2 1N4148W to CLAMP_SW. Values marked TBD are set with the part choices (design-log.md).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Antenna port A3/
File: antenna_port.kicad_sch

Title: Unit RF board: antenna port (K1, driver, detector, bias-T)

Size: A4	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 8/13



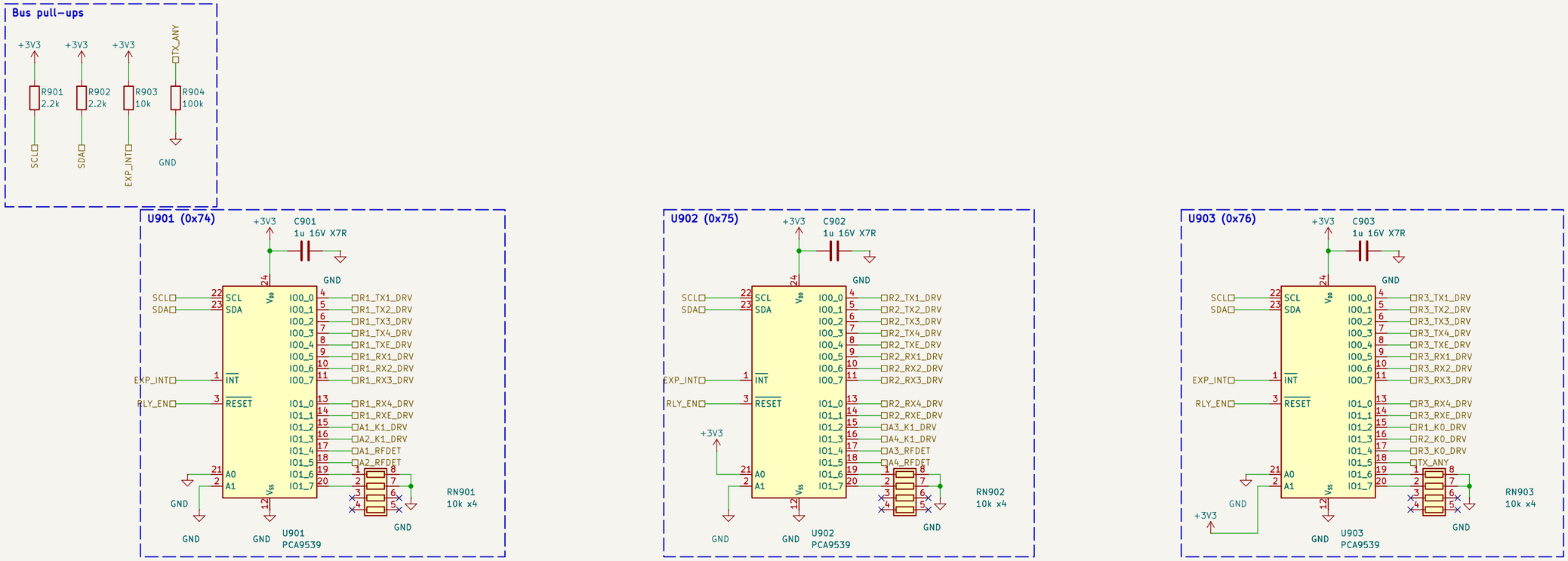
Antenna port (one sheet, used for A1 to A4; references 4xx to 7xx). SO-239: root sheet.
K1: coil off = antenna parked (COM to NC, 1M bleed to GND), coil on = antenna on the column.
No RF choke on the bleed: a low-impedance path would take surge current through K1 and the board.
Surge: Comet CS-400R at the RVS box entry. No GDT on the board (D-114).
Bias-T option (A2 RX loop): fit C1 and L2, C2, remove R2. The feed sits on the matrix side of K1,
so the bias supply is open, not loaded, while K1 is parked.
Detector: capacitive tap C4/C5 (about 1:12), BAS70-04 doubler D1 (clamps negative pulses), hold C3, R5/R4 divide by 2.
K1 driver: Q1 BSS138BK, R6 100k gate pull-down, D2 1N4148W to CLAMP_SW. Values marked TBD are set with the part choices (design-log.md).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Antenna port A4/
File: antenna_port.kicad_sch

Title: Unit RF board: antenna port (K1, driver, detector, bias-T)

Size: A4	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 9/13



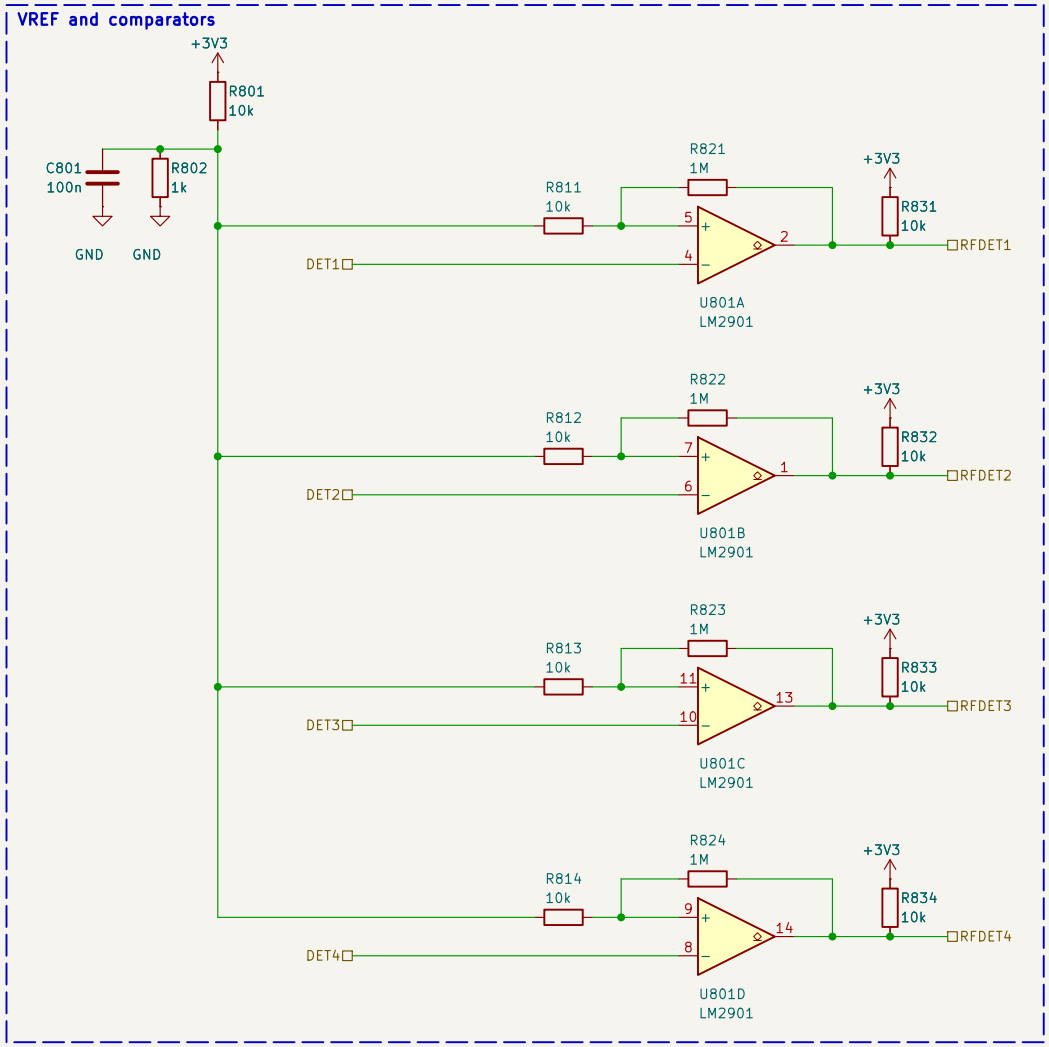
I/O expanders (D-49, D-50): U901 0x74, U902 0x75, U903 0x76 (A1 A0 = 00, 01, 10); U1001 0x77 on the bias-T sheet (D-54).
Same addresses on unit 2,
which runs on its own I2C channel from the controller. Outputs drive the coil MOSFET gates (active high, 100k pull-downs
at each gate), so power-on, RESET and a lost bus leave every coil off. RESET = RLY_EN (active high, R3 pull-down on
the power sheet): no signal holds the expanders in reset and +12V_SW off. Documented exception: the datasheet asks
for a pull-up on RESET. Firmware: set RLY_EN high, then configure. Spare I/Os: 10k to GND in RN901 to RN903 (D-121).
EXP_INT (open drain) signals a change on the RFDET, R1_K0_DRV and TX_ANY (PTT readback, D-53) inputs.
U903 IO1_3 and IO1_4 are outputs: they drive R2 and R3 K0 after the TX handshake (D-63). In reset they float: the 100k gate pull-downs keep K0 off.
Bus pull-ups on this board, so unit 1 and unit 2 are the same (D-52): R901, R902 2.2k on SCL, SDA (bus about 100 pF max:
rise 0.85 x 2.2k x 100 pF = 0.19 us, Fast-mode limit 0.3 us; sink 1.5 mA, PCA9539 IOL min 3 mA). R903 10k on EXP_INT.
The controller board adds no pull-ups on this bus. Unit 2: J14 takes the EXP ribbon from the unit 1 controller board (D-53).
R904 100k pull-down on TX_ANY: defined low with J14 open (logic verification F4, D-61).

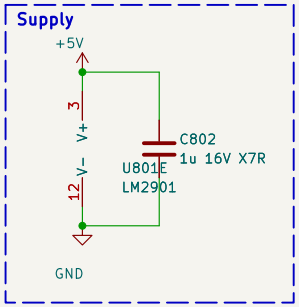
JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /IO expanders/
File: io_expanders.kicad_sch

Title: Unit RF board: I/O expanders (3 x PCA9539)

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 10/13





RF detect: one LM2901 for the four antenna ports (D−47). DETk from antenna port Ak (BAS70 peak detector).
Threshold VREF 0.3 V (R801/R802); DET above VREF pulls RFDETk low (active low, open collector, 10k pull−up to 3.3 V).
Hysteresis about 33 mV per channel (R81k 10k series, R82k 1M feedback). Supply +5V: common−mode range to VCC − 1.5 V;
inputs rated to 36 V, DET max about 15 V at 100 W SWR 3. Detects about 0.5 W (SWR 1) to 2.5 W (SWR 3, voltage minimum).
RFDET1 to RFDET4 go to the expander inputs. RFDET2 (A2, receive−only) also goes to J14 pin 17: the controller opens
the SAFE loop on RF at A2, so the shack inhibits all radios and the relays drop (D−104, SR−01 H2).

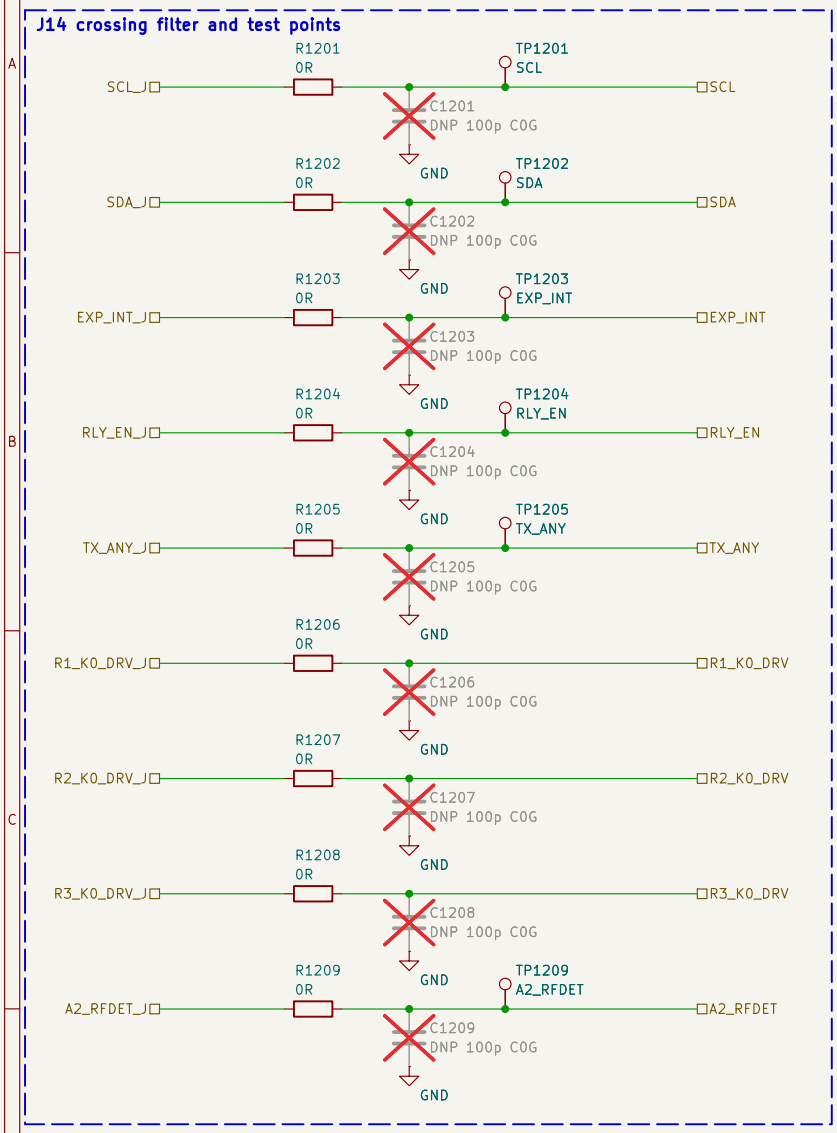
JLC06161H−7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO−239 Amphenol 83−798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

PE1RGU − Shack Relay Matrix

Sheet: /RF detect/
File: rf_detect.kicad_sch

Title: Unit RF board: RF detect comparators (LM2901)

Size: A4	Date: 2026−10−22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 11/13



J14 crossing (D-122): each slow line from the controller board enters through a series link (0R fitted) with a capacitor footprint to GND (not fitted), placed at J14. Values are set on the bench: series 100R to 1k, C up to 100 pF; check the I2C rise time with the 2.2k pull-ups (IO sheet). Test points on the logic side (D-120). Nets *_J run from J14 to this sheet only.

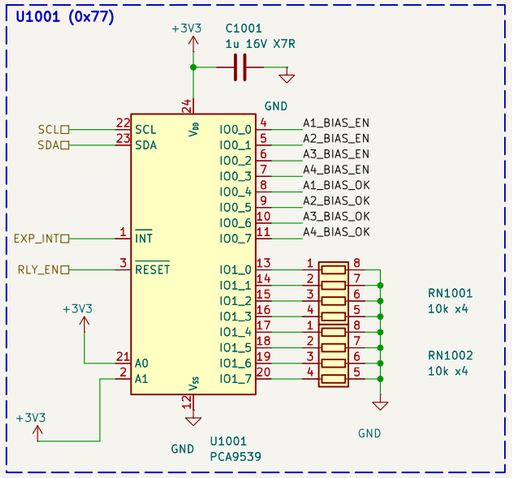
JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU

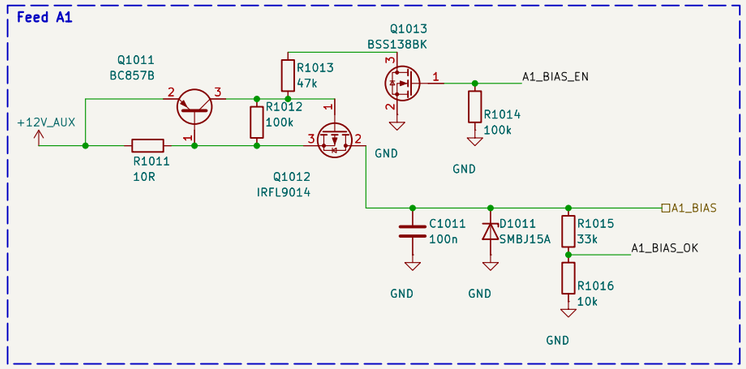
PE1RGU — Shack Relay Matrix

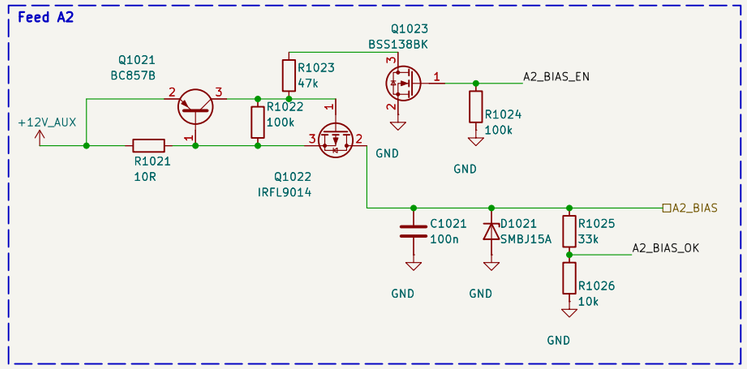
Sheet: /J14 filter/
File: j14_filter.kicad_sch

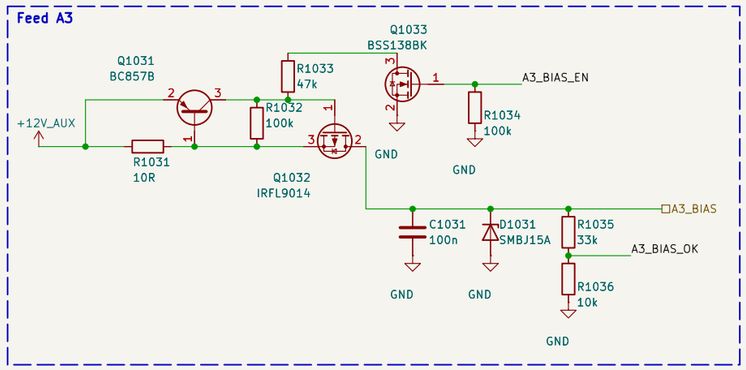
Title: Unit RF board: J14 crossing filter and test points

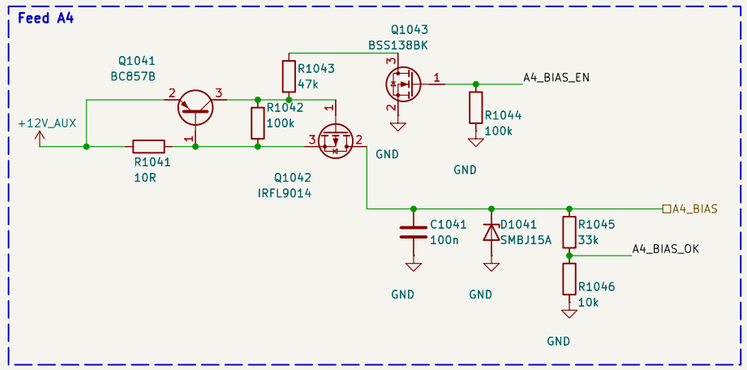
Size: A4	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 12/13











Bias-T supply (D-54). U1001 PCA9539 at 0x77 (A1 A0 = 11), on the same bus and RESET (= RLY_EN) as U901 to U903.
IO0_0..3 = A1..A4_BIAS_EN (outputs, active high). IO0_4..7 = A1..A4_BIAS_OK (inputs). IO1 spare, held low by RN1001, RN1002 (D-121).
Per feed: +12V_AUX, 10R sense, IRFL9014 P-MOSFET high side, BC857B limiter (about 42 to 78 mA over temperature), BSS138BK gate drive.
A short on the coax holds the feed in current limit: 0.8 W max in the P-MOSFET, Tj about 103 C at 55 C ambient on 1 sq. inch copper (C-17).
The firmware switches a feed off when EN is high and OK stays low (fault), and reports it. OK changes raise EXP_INT.
Safe state: RLY_EN low holds U1001 in reset, the 100k pull-downs keep every feed off (KILL also turns the preamps off).
Ak_BIAS goes to the bias-T of antenna port Ak (L2, C2 DNP there; fit on a bias-T port only).

JLC06161H-7628A, 1.6 mm, 6 layers. RF: CPWG W 0.975 / gap 2.0 mm, 50 ohm
Crossbar: rows on L1, columns on L6. Unit 1 = 3 radios x 4 antennas + EXP
RT1 RTD14012F x37, SO-239 Amphenol 83-798 x7, SMA EXP x3, XT60 x2, XT30
Design: Prins Doornekamp, PE1RGU
PE1RGU - Shack Relay Matrix

Sheet: /Bias-T supply/
File: bias_supply.kicad_sch

Title: Unit RF board: bias-T supply (switched, current-limited)

Size: A3	Date: 2026-10-22	**Rev: C**
KiCad E.D.A. 10.0.5		Id: 13/13